

accessible estimator states. Parameter mismatch is excluded from the common cross-model ranking and is examined through the supplementary HECM analysis defined in Section 6.2.7. Figure 6.2 groups the scenarios into input disturbances, initialization errors, and signal-integrity faults. Table 6.2 specifies their implementation.[9]

The disturbance magnitudes represent deployment-level sensing limits, adverse measurement corruption, and signal-integrity stress conditions.[19, 21, 23, 22, 117, 118] Random sensor noise models zero-mean fluctuations in current, voltage, and temperature. Systematic sensor errors comprise multiplicative current-gain error and additive offsets in current, voltage, and temperature. Voltage spikes and ADC quantization complete the input-disturbance branch. The remaining scenarios address initialization, measurement availability, and sampling-time integrity.

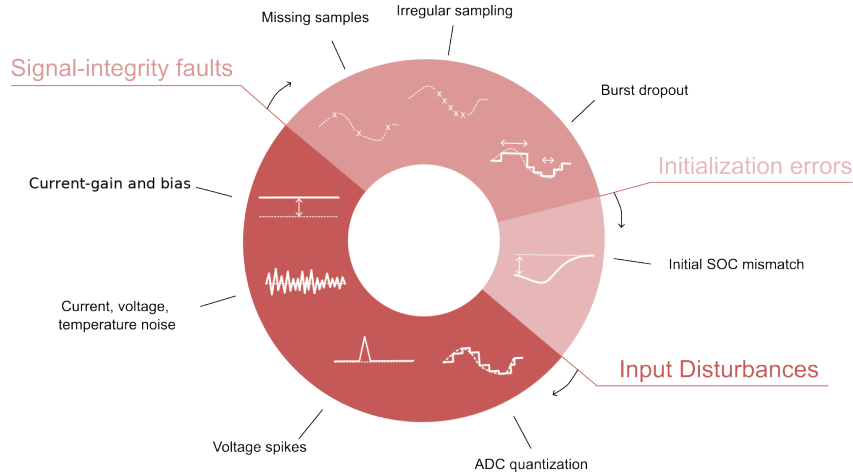


Figure 6.2: Taxonomy of the benchmark scenarios, grouped into input disturbances, initialization errors, and signal-integrity faults.

Table 6.2: Disturbance configurations and practical rationale. The numerical values are fixed benchmark settings, while the cited sources motivate the disturbance mechanisms and magnitude ranges.

Scenario	Benchmark setup	Practical origin and rationale
ADC quantization	$\Delta I = 0,01 \text{ A}$, $\Delta U = 0,005 \text{ V}$, and $\Delta T = 0,5 \text{ }^\circ\text{C}$	ADC steps, sensor scaling, rounding, and coarse deployment-level discretization.[19, 21, 20]